

	Week 1					Week 2				
	Day 1	Day 2	Day 3	Day 4	Day 5	Day 6	Day 7	Day 8	Day 9	Day 10
9:00 – 10:00	Background 1 What are SDMs and why we use them.	Data Download Download GBIF and climate data for the teacher example.	Run First SDM GAMs for the teacher example.	Model Projection Where to find future climate data	Theory Advanced What are we modelling with SDMs? What are the limitations of SDMs? Group discussion.	Individual project				Student Presentations Explain the case study. Illustrate the workflow. Show the results. Group feedback.
10:00 – 11:00	Theory Primer The basic theory behind SDMs and how they work.	GIS Primer extract(), spatSample(), buffer(), etc.		Model Projection Temporal projection.						
11:00 – 12:00										
12:00 – 13:00										
13:00 – 14:00	Stats GAMs	GIS Primer extract(), spatSample(), buffer(), etc.	Practical 1 Model selection and interpretation	<i>Free flexible Slot</i>	Data Download Students download their own GBIF and climate data + set up their workflow	Teacher Feedback				<i>Free flexible slot</i>
14:00 – 15:00		Data Processing Prepare data for	Model Projection	Background 2 Why and for what people have used SDMs (two						<i>Free flexible slot</i>

15:00 – 16:00	Data Where to find data to run a SDM? GBIF + WorldClim.	Prepare data for the teacher example.	Spatial projection.	SDMs (two examples, one for one species regionally and one macroecol.)	Workflow.	Wrap Up + Assignment
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Spatial and temporal projection + where to find future climate data.

Species distribution models

This course introduces the theory and practice of Ecological Niche Modelling (ENM) and Species Distribution Modelling (SDM) and their applications in ecology and conservation. Students will learn the core concepts, data requirements, and basic modelling approaches, supported by hands-on practical sessions using real-world datasets. By the end of the course, students will be able to independently build, evaluate, and interpret basic ENM/SDM analyses.

Getting
familiar with
GIS

Running
SDMs

Data